

Interaction of Calcium with Native and Decarboxylated Human Factor X.

EFFECT OF PROTEOLYSIS IN THE AUTOLYSIS LOOP ON CATALYTIC EFFICIENCY AND FACTOR Va BINDING*

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Human factor X is a two-chain, 58-kDa, vitamin K-dependent blood coagulation zymogen. The light chain of factor X consists of an NH₂-terminal γ -carboxyglutamic acid (Gla) domain, followed by a few helical hydrophobic residues and the two epidermal growth factor-like domains, whereas the heavy chain contains the serine protease domain. In this study, native factor X was found to contain three classes of Ca²⁺-binding sites: two high affinity (K_d 100 $\pm$ 30 μ M), four intermediate affinity (K_d 450 $\pm$ 70 μ M), and five to six low affinity (K_d 2 $\pm$ 0.2 mM). Decarboxylated factor X in which the Gla residues were converted to Glu retained the two high affinity sites (K_d 140 $\pm$ 20 μ M). In contrast, **factor X lacking the Gla domain as well as a part of the helical hydrophobic residues (des-44-X) retained only one high affinity Ca²⁺-binding site (K_d 130 $\pm$ 20 μ M).** Moreover, **a synthetic peptide composed of residues 238–277 (58–97 in chymotrypsinogen numbering) from the protease domain of factor X bound one Ca²⁺ with high affinity (K_d 150 $\pm$ 20 μ M).** From competitive inhibition assays for binding of active site-blocked factor Xa to factor Va in the prothrombinase complex, the K_d for peptide-Va interaction was calculated to be $\sim$ 10 μ M as compared with 30 μ M for factor Xa and $\sim$ 1.5 μ M for decarboxylated factor Xa. A peptide containing residues 238–262(58–82) bound Ca²⁺ with reduced affinity (K_d $\sim$ 600 μ M) and did not inhibit Xa:Va interaction. In contrast, a peptide containing residues 253–277(73–97) inhibited Xa:Va interaction (K_d $\sim$ 10 μ M) but did not bind Ca²⁺. In additional studies, Ca²⁺ increased the amidolytic activity of native and des-44-Xa toward a tetrapeptide substrate (benzoyl-Ile-Glu-Gly-Arg-*p*-nitroanilide) by approximately 1.6-fold. The half-maximal increase was observed at $\sim$ 150 μ M Ca²⁺ and the effect was primarily on the k_{cat} . Ca²⁺ also significantly protected cleavage at Arg-332-Gln-333(150–151) in the protease domain autolysis loop. Des-44-Xa in which the autolysis loop was cleaved possessed 5% of the amidolytic activity of the noncleaved form; however, the S1 binding site was not affected, as determined by the *p*-aminobenzamidine binding. Additionally, autolysis

loop-cleaved, active site-blocked native factor Xa was calculated to have $\sim$ 10-fold reduced affinity for factor Va as compared with that of the noncleaved form.

Factor X is a vitamin K-dependent multidomain protein that participates in the middle phase of blood coagulation (1). Factor X is essential for hemostasis since a reduction in its functional activity results in a rare autosomal recessive bleeding disorder known as Stuart-Prower factor deficiency (2). The human protein is synthesized in the liver as a precursor molecule of 488 amino acids (3). The amino-terminal 40 amino acids constitute the prepro leader sequence, which is removed prior to secretion of the molecule. Additionally, during biosynthesis, the protein undergoes several posttranslational modifications including glycosylation, γ -carboxylation (of the first 11 glutamic acid residues), β -hydroxylation (of Asp-63), and removal of a tripeptide (Arg-Lys-Arg) between Arg-139 and Ser-143 (3). The resulting mature protein is a zymogen of serine protease factor Xa and consists of a light chain (amino acids 1–139) and a heavy chain (amino acids 143–448) held together by a single disulfide bond between Cys-132 and Cys-302.

Gene arrangement, amino acid sequence, and modular structure of factor X strongly suggest that the protein is organized into several distinct domains (3, 4). The amino terminus of the light chain contains 11 γ -carboxyglutamic acid (Gla)¹ residues and represents the Gla domain (residues 1–39) of factor X (3). The Gla domain is followed by a short hydrophobic stack (residues 40–45) and two epidermal growth factor (EGF)-like domains (residues 46–84 (EGF1) and residues 85–128 (EGF2)). The amino terminus of the heavy chain of factor X contains the activation peptide region of 52 amino acids (residues 143–194) followed by the serine protease domain of 254 amino acids (residues 195–448), which features the active site triad of His-236(57),² Asp-282(102), and Ser-379(195) (3).

¹ The abbreviations and trivial names used are: Gla, γ -carboxyglutamic acid; DEGR-CK, dansyl-Glu-Gly-Arg-chloromethyl ketone; EGF, epidermal growth factor; des-44-X or -Xa, Gla-domainless human factor X or Xa; PC, phosphatidylcholine; PS, phosphatidylserine; BSA, bovine serum albumin; PL, phospholipid; *p*-AB, *p*-aminobenzamidine; peptide 1, peptide containing the sequence of 238–277(59–97) residues of factor X; peptide 2, peptide containing the sequence of 238–262(58–82) of factor X; peptide 3, peptide containing the sequence of 253–277(73–97) of factor X; S-2222, benzoyl-Ile-Glu-Gly-Arg-*p*-nitroanilide; S-2238, H-D-Phe-Pip-Arg-*p*-nitroanilide; DEGR-Xa, factor Xa inactivated with DEGR-CK; DEGR-des-44-Xa, des-44-Xa inactivated with DEGR-CK; TBS, Tris-buffered saline; IIa, α -thrombin; dansyl, 5-dimethylaminonaphthalene-1-sulfonyl.

² For comparison, factor X amino acid numbering system has been used. The number in parentheses refers to the chymotrypsin equivalents for the protease domain of factor Xa.

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During physiologic hemostasis, factor X can be activated by factor IXa, requiring Ca^{2+} , phospholipid (PL), and factor VIIIa or factor VIIa requiring Ca^{2+} and tissue factor (1, 5). A potent nonphysiologic activator of factor X is the coagulant protein from Russell's viper venom (6). In all cases, the activation results from the cleavage of the Arg-194-Ile-195(15-16) bond in the heavy chain of factor X and release of a 52-residue activation peptide; the light chain remains unaltered during this process (3, 7). Factors X and $\text{Xa}\alpha^3$ can also be converted to their respective β -forms where ~ 4 kDa peptide is cleaved off from the COOH terminus of the heavy chain; this, however, does not result in a loss of coagulant activity (6). Factor Xa converts prothrombin to thrombin in the coagulation cascade; for a physiologically significant rate, this reaction requires Ca^{2+} , PL, and factor Va (8). Thus, Ca^{2+} plays an important role both in the activation of factor X and in the conversion of prothrombin to thrombin by factor Xa.

Ca^{2+} binding to human factor X has been studied by Monroe *et al.* (9). The authors reported that the protein contains one high affinity Gla-independent and 19 weak Gla-dependent Ca^{2+} -binding sites (9). In the present report, we have extensively investigated the Ca^{2+} -binding properties of native, decarboxylated, and Gla domainless (des-44-X) human factor X. The data strongly indicate that the Gla domain contains four intermediate and five to six low affinity Ca^{2+} -binding sites, whereas the EGF1 and the protease domains each contains one high affinity Ca^{2+} -binding site. Further, proteolysis in the autolysis loop of the catalytic domain results in a virtual loss of amidolytic activity without affecting the S1 binding site. Autolysis loop-cleaved factor Xa ($\text{Xa}\gamma$) also has ~ 10 -fold reduced affinity for factor Va. Importantly, Ca^{2+} protects the proteolytic cleavage in the autolysis loop, thereby stabilizing this domain for maximal biologic activity. An initial account of this work has been presented in abstract form (10).

EXPERIMENTAL PROCEDURES

Proteins and Reagents— $^{45}\text{CaCl}_2$ (7.4 Ci/g of calcium) and $^3\text{H}_2\text{O}$ (5 Ci/g) were obtained from ICN. Benzoyl-Ile-Glu-Gly-Arg-*p*-nitroanilide (S-2222) and H-D-Phe-Pip-Arg-*p*-nitroanilide (S-2238) were purchased from Helena Laboratories. Dansyl-Glu-Gly-Arg-chloromethyl ketone (DEGR-CK) was obtained from Calbiochem. *N* $^{\alpha}$ -*p*-Tosyl-L-lysine chloromethyl ketone-treated α -chymotrypsin, diisopropyl fluorophosphate, *p*-aminobenzamide (*p*-AB), polyethylene glycol 8000, bovine serum albumin (BSA), bovine brain phosphatidylcholine (PC), and phosphatidylserine (PS) were obtained from Sigma. Human factor X and prothrombin were purified as described earlier (11). A total of ~ 100 mg of factor X obtained from six isolations, each starting with 8 liters of plasma, was used for these investigations. Decarboxylated factor X was prepared as outlined previously (12). Five preparations of 10 mg each were made and pooled. Small amounts of higher molecular weight materials ($\sim 10\%$ of the total protein) in decarboxylated factor X were removed by gel filtration utilizing a Sephadex G-100 column (2.5×150 cm) equilibrated in TBS, pH 7.4 (0.05 M Tris, 0.15 M NaCl, pH 7.4). Des-44-X was prepared and purified by QAE-Sephadex A-50 column chromatography as described by Morita and Jackson (13). The NH_2 -terminal sequence analysis of des-44-X revealed two sequences of approximately equimolar amounts, one corresponding to the heavy chain (Ser-Val-Ala-Gln-Ala) and the other corresponding to the modified light chain, des-Gla L (Lys-Asp-Gly-Asp-Gln). Decarboxylated factor X and des-44-X each contained <0.5 Gla/mol as compared with factor X, which contained 10.7 ± 0.6 Gla/mol as measured by the specific ^3H incorporation (14). Factor $\text{Xa}\beta$, des-44- $\text{Xa}\beta$, and decarboxylated $\text{Xa}\beta$ were prepared using insolubilized Russell's viper venom as outlined previously (11, 13). PCPS vesicles (75% PC, 25% PS) were prepared as earlier

(15). Human factor Va was obtained from Hematologic Technologies, Inc., and α -thrombin (IIa) was purchased from Enzyme Research Laboratories, Inc.

Des-44- $\text{Xa}\gamma$ was prepared by incubation of 10 mg (2 mg/ml) of des-44- $\text{Xa}\beta$ in TBS, pH 7.4, containing 1 mM EDTA for 30 h at 37 °C. SDS-gel electrophoretic analysis revealed that after a 30-h incubation period, all of the H β had been converted to H γ N and H γ C (see "Results and Discussion"). In our efforts to prepare native factor $\text{Xa}\gamma$, we found that in addition to proteolysis in the autolysis loop, the Gla domain was also cleaved off, albeit slowly.⁴ Therefore, to investigate the properties of factor $\text{Xa}\gamma$, we prepared a mixture of factors $\text{Xa}\beta$ and $\text{Xa}\gamma$ (factor $\text{Xa}\beta\gamma$) as follows. We incubated factor $\text{Xa}\beta$ in TBS, pH 7.4, in the presence of 1 mM EDTA for 5 h at 37 °C. This sample, as analyzed by reduced SDS gels, contained both factor $\text{Xa}\beta$ and factor $\text{Xa}\gamma$ as well as des-44- $\text{Xa}\beta$ and des-44- $\text{Xa}\gamma$. Des-44- $\text{Xa}\beta\gamma$ was removed from factor $\text{Xa}\beta\gamma$ utilizing Mono Q fast protein liquid chromatography as outlined for separating Gla-domainless factor VIIa from native factor VIIa (17). The purified factor $\text{Xa}\beta\gamma$ was free of Gla-domainless factor $\text{Xa}\beta\gamma$ (see "Results and Discussion").

DEGR- $\text{Xa}\beta$, DEGR-des-44- $\text{Xa}\beta$, and decarboxylated DEGR- $\text{Xa}\beta$ were prepared by incubating the enzymes (~ 500 $\mu\text{g}/\text{ml}$) in TBS, pH 7.4, with 20-fold molar excess of DEGR-CK for 2 h at 37 °C, at which time an additional 20-fold excess of the inhibitor was added to each tube and the pH adjusted to 7.4. The samples were then allowed to sit overnight at 4 °C, and the excess inhibitor in each sample was removed as described earlier (15). The preparations had no measurable S-2222 hydrolytic activity. DEGR-des-44- $\text{Xa}\gamma$ and DEGR- $\text{Xa}\beta\gamma$ were prepared as above, except three successive additions of the DEGR-CK were made instead of the two earlier; after the second addition of the inhibitor, each tube was incubated for 2 h at 37 °C prior to the last addition and incubation overnight. When a known extinction coefficient (3940 M^{-1} at 340 nm) of the dansyl probe (18) was used, we obtained stoichiometric (1.1 ± 0.05) incorporation of the inhibitor into each factor Xa protein.

Ca^{2+} -binding Studies—Binding of Ca^{2+} to factor X, decarboxylated factor X, and des-44-X was investigated by the technique of equilibrium dialysis using ^{45}Ca , as described in detail previously (19). The only change from the described method was the use of microcells of 0.1 ml volume. At the end of each experiment, the proteins were analyzed by SDS-gel electrophoresis and were stable for at least 48 h at room temperature. Ca^{2+} binding to the peptides was determined using a Ca^{2+} -specific electrode and a model 601A digital ion analyzer. Titrations of peptides in 4 ml of buffer were performed by adding small increments (4–8 μl) of 10 or 100 mM CaCl_2 at room temperature. In these titrations, bound Ca^{2+} was taken as the difference between measured free Ca^{2+} concentration and total added (15). Data were analyzed using the nonlinear, least-squares, curve fitting program LIGAND (20).

Peptide Synthesis—A total of three peptides were synthesized using Merrifield's solid phase method (21) on an Applied Biosystems model 430A peptide synthesizer. The peptides were deprotected and cleaved from the resin with anhydrous hydrogen fluoride/anisole/dimethyl sulfide (10:1:1) (v/v/v) for 50 min at 0 °C. The cleaved peptides were washed with diethyl ether and extracted from the resin with 30% acetic acid. After removing the acetic acid by rotary evaporation, the remaining aqueous solution was diluted 4-fold with water, shell frozen, and lyophilized. Peptides were further purified ($\geq 90\%$) by reverse phase high performance liquid chromatography on a Vydac C-18 (22×250 mm) column using standard trifluoroacetic acid/acetonitrile conditions (22). The sequence of peptide 1 was: Leu-Tyr-Gln-Ala-Lys-Arg-Phe-Lys-Val-Arg-Val-Gly-Asp-Arg-Asn-Thr-Glu-Gln-Glu-Glu-Gly-Gly-Ala-Val-His-Glu-Val-Glu-Val-Val-Ile-Lys-His-Asn-Arg-Phe-Thr-Lys-Glu. The sequence of peptide 1 corresponds to residues 238–277(58–97) of human factor X (3, 4). The sequence of peptide 2 corresponds to residues 238–262(58–82), and the sequence of peptide 3 corresponds to residues 253–277(73–97) of factor X. Tyrosine was added at the NH_2 terminus of peptide 3 to facilitate determination of its concentration in solution. Peptide concentrations were determined using the molar extinction coefficient of 2390 at 293 nm for tyrosine in 0.1 M NaOH (23).

SDS-Gel Electrophoresis—SDS-gel electrophoresis was performed using the Laemmli buffer system (24). The acrylamide concentration was 15%, and the gels were stained with Coomassie Brilliant Blue dye. All proteins used in the present study were $\sim 98\%$ pure.

³ The nomenclature used for factor Xa is that of Di Scipio *et al.* (6). Factor $\text{Xa}\alpha$ is factor Xa consisting of a light chain (L, amino acids 1–139) and a heavy chain (H α , amino acids 195–448). Factor $\text{Xa}\beta$ is factor $\text{Xa}\alpha$ lacking the ~ 4 kDa fragment from the COOH terminus of the H α chain. Factor $\text{Xa}\gamma$ is factor $\text{Xa}\beta$ in which proteolysis has occurred in the autolysis loop in the H β chain. Factor $\text{Xa}\beta\gamma$ is a mixture of factors $\text{Xa}\beta$ and $\text{Xa}\gamma$.

⁴ We also attempted to prepare factor $\text{Xa}\gamma$ by treatment of factor $\text{Xa}\beta$ with plasmin in the presence of Ca^{2+} and PL as recently described by Prydzial and Kessler (16). In our experiments, factor $\text{Xa}\gamma$ formation was transient and it was rapidly degraded to small molecular weight fragments ranging from 11,000 to 18,000 in reduced SDS gels.

Molecular Modeling—The putative model of the Gla domain of factor X was constructed using a homology model building approach described earlier (25). Crystallographic structures of the Gla domain of prothrombin in the presence of Ca^{2+} (26) and Sr^{2+} (27) were used as the starting templates. EGF1 domain was modeled using the NMR coordinates of EGF1 domain of bovine factor X in the presence of Ca^{2+} (28). Structures of the EGF2 and protease domains of human factor Xa were from Padmanabhan *et al.* (4). In this structure, Glu-250(80) side chain was disordered and there was no electron density beyond the β -carbon; the side chain of this glutamic acid was introduced during the modeling of Ca^{2+} -binding site in the 250–260(70–80) loop. A single Ca^{2+} -binding site in trypsin was first identified by Bode and Schwager (29); coordinates of the Ca^{2+} -binding loop were taken from the Protein Data Bank (Brookhaven National Laboratory, code 4PTP) for modeling of the Ca^{2+} -binding site in factor Xa. In modeling of the whole factor Xa molecule, individual domains with respect to each other were positioned based upon the crystal structure of porcine factor IXa (Ref. 30, code 1PFX), a protein whose domain organization and modular structure is similar to that of factor Xa.

Amino Acid Sequence Analysis—Automated Edman degradation of each protein component was performed using an Applied Biosystems model 477A gas phase sequencer. Approximately 0.1–0.5 nmol of protein was loaded on the filter cartridge. The proteins from SDS gels were transferred to polyvinylidene difluoride membranes as described by Rosenberg (31).

Measurements of S-2222 Amidolytic Activity of Factor Xa Proteins—The concentration of factor Xa β and des-44-Xa β used was 1 nM each and the concentration of des-44-Xa γ was 20 nM. The S-2222 concentration ranged from 20 μM to 1 mM. The buffer used was TBS, 0.1 mg/ml BSA, pH 7.4, containing 1 mM Ca^{2+} or 1 mM EDTA. The *p*-nitroaniline release was measured continuously ($\Delta A_{405}/\text{min}$) for up to 30 min using a Beckman DU65 spectrophotometer equipped with a Soft-Pac kinetics module. An extinction coefficient of 9.9 $\text{mM}^{-1}\text{cm}^{-1}$ at 405 nm was used in calculating the amount of *p*-nitroaniline released (32). All reactions were performed in triplicate. The K_m and k_{cat} values were obtained using the Enzyme Kinetics program from Erithacus Software. In Ca^{2+} titration experiments, the Ca^{2+} concentrations ranged from 0–500 μM (0, 20, 50, 100, 200, 300, 400, and 500 μM), and the substrate concentration was 170 μM .

***p*-AB Binding**—Binding of *p*-AB was measured by increase in its intrinsic fluorescence upon binding to the active site of each factor Xa protein using Perkin-Elmer 650-10S Fluorescence Spectrophotometer. The concentration of each factor Xa protein used was 100 $\mu\text{g}/\text{ml}$ (1.7 μM) in TBS, pH 7.4 containing 0.3% polyethylene glycol 8000, and the excitation wavelength was 336 nm (33). A titration of the protein solution (700 μl) in the presence of EDTA (1 mM) or Ca^{2+} (5 mM) was performed by adding small increments (2–4 μl) of 1.8 mM stock solution of *p*-AB, and the resulting fluorescence at 376 nm was measured at each point after the attainment of equilibrium conditions (usually 1 min). Both excitation bandwidth and emission bandwidth were 5 nm.

Measurements of K_d Values for the Interaction of Each Factor Xa Species with Factor Va in the Prothrombinase Complex—For these experiments, we first determined the functional K_d (EC_{50}) of interaction of factor Xa β with factor Va in our system. Reaction mixtures (50 μl final volume) were prepared in Eppendorf tubes containing 5 pM factor Va, 10 μM PCPS, 5 mM Ca^{2+} , and varying concentrations (5, 12, 19, 25, 37, 50, 75, and 100 pM) of factor Xa β . The buffer used was TBS/BSA, pH 7.4 (TBS containing 1 mg/ml BSA). After 5 min at 37 $^\circ\text{C}$ prothrombin (700 nM, final concentration) was added to each tube, and the mixture was incubated for a variable time period ranging from 2 to 10 min. A 20- μl aliquot of the reaction mixture was removed at a given time and added to 10 μl of 30 mM EDTA to stop further generation of thrombin. Twenty μl of each chelated sample was then transferred to a 0.1-ml quartz cuvette containing 100 μl of S-2238 (final concentration, 125 μM). The *p*-nitroaniline release was measured continuously ($\Delta A_{405}/\text{min}$) for up to 30 min. All reactions were performed in triplicate. Thrombin generated was calculated from a standard curve constructed using purified thrombin. Functional K_d was obtained from a plot of factor Xa *versus* rate of thrombin generation using the Enzyme Kinetics program from Erithacus Software. The functional K_d of factor Xa β :Va interaction at several additional concentrations (15, 50, 150, 300, 500, 1200, and 1500 nM) of prothrombin was also determined as outlined above using 750 nM prothrombin. K_d values of factor Va binding to DEGR-Xa $\beta\gamma$, of decarboxylated DEGR-Xa β , and of the three peptides were determined from their abilities to inhibit prothrombin activation in a system containing 5 pM factor Va, 15 pM factor Xa β , 10 μM PCPS, and 750 nM prothrombin. In each case factor Xa β was mixed with the competitor

prior to incubation with factor Va. The steady state inhibition curves were analyzed using the program LIGAND (20).

RESULTS AND DISCUSSION

Ca^{2+} -binding Studies—SDS-gel electrophoretic analysis of the proteins used in Ca^{2+} -binding experiments is presented in Fig. 1A. Under nonreducing conditions, each of the three proteins (native factor X, decarboxylated factor X, and des-44-X) used for Ca^{2+} -binding experiments revealed a single Comassie Brilliant Blue dye staining band. Under reducing conditions, native and decarboxylated factor X revealed the typical heavy and light chains, whereas the des-44-X protein revealed a smaller light chain indicative of removal of the Gla domain. The NH_2 -terminal sequence analysis of des-44-X is given under “Experimental Procedures.” In addition, the disulfide reduced native factor X contained a small amount of high molecular weight component that could represent an incompletely processed protein in which the tripeptide (residues 140–142) has not been removed (3, 7). An apparent absence of this high molecular weight component in decarboxylated factor X may indicate its removal during the gel filtration step (see “Experimental Procedures”). The purity of the factor X preparations is shown here, since it is important to the conclusions drawn from the data presented in this paper.

The Scatchard plots of the Ca^{2+} -binding data for native factor X and decarboxylated factor X as measured by the equilibrium dialysis technique are shown in Fig. 1B. The data on native factor X fit to a model in which there are three classes of Ca^{2+} -binding sites: two high affinity ($K_d \sim 100 \mu\text{M}$), four intermediate affinity ($K_d \sim 450 \mu\text{M}$), and five to six low affinity ($K_d \sim 2 \text{ mM}$). An analysis by the LIGAND program (20) gave a *p* value (significance of fit) of <0.01 between the high and intermediate affinity sites and of <0.05 between the intermediate and low affinity sites. The two high affinity sites are present in the non-Gla region of the molecule since decarboxylated factor X retains these sites (Fig. 1B, *inset*). In contrast, des-44-X contained only one high affinity site (data not shown). Ca^{2+} -binding data are summarized in Table I. From the structural similarities and identical domain organizations between factors VII, IX, and X, it is clear that out of the two non-Gla high affinity sites in factor X, one is located in the EGF1 domain (15, 34, 35) and the other in the protease domain (10, 36–39). Because des-44-X has only one high affinity Ca^{2+} -binding site (Ref. 9 and the present study) as compared with the two in decarboxylated factor X, it is likely that residues Asp-46 to Gln-49 are flexibly disordered in des-44-X and cannot effectively participate in forming the high affinity site in the EGF1 domain (40).

Since decarboxylated factor X retained the two high affinity Ca^{2+} -binding sites, it is reasonable to conclude that the intermediate and low affinity sites in native factor X are located in the Gla region of the molecule. Bovine factor X (41) and prothrombin (42–44) have been reported to bind Ca^{2+} with positive cooperativity and two or three sites may be involved in this interaction; furthermore, the possibility of four cooperative sites may not be ruled out (45). In light of these observations, we explored whether or not the intermediate affinity sites in the Gla domain of human factor X exhibited positive cooperativity in binding to Ca^{2+} . As is evident from Fig. 1 (B–D), we were unable to demonstrate cooperative binding of Ca^{2+} to native (Hill coefficient 0.87) or decarboxylated (Hill coefficient 0.91) factor X from the direct measurement data. However, cooperativity in Ca^{2+} interaction became evident when we obtained a hypothetical curve of binding to the Gla domain by subtracting the binding curve of the decarboxylated factor X from that of the native factor X (Fig. 1C). The Hill coefficient, at midpoint of 0.1 mM Ca^{2+} concentration was 1.57 and at mid-

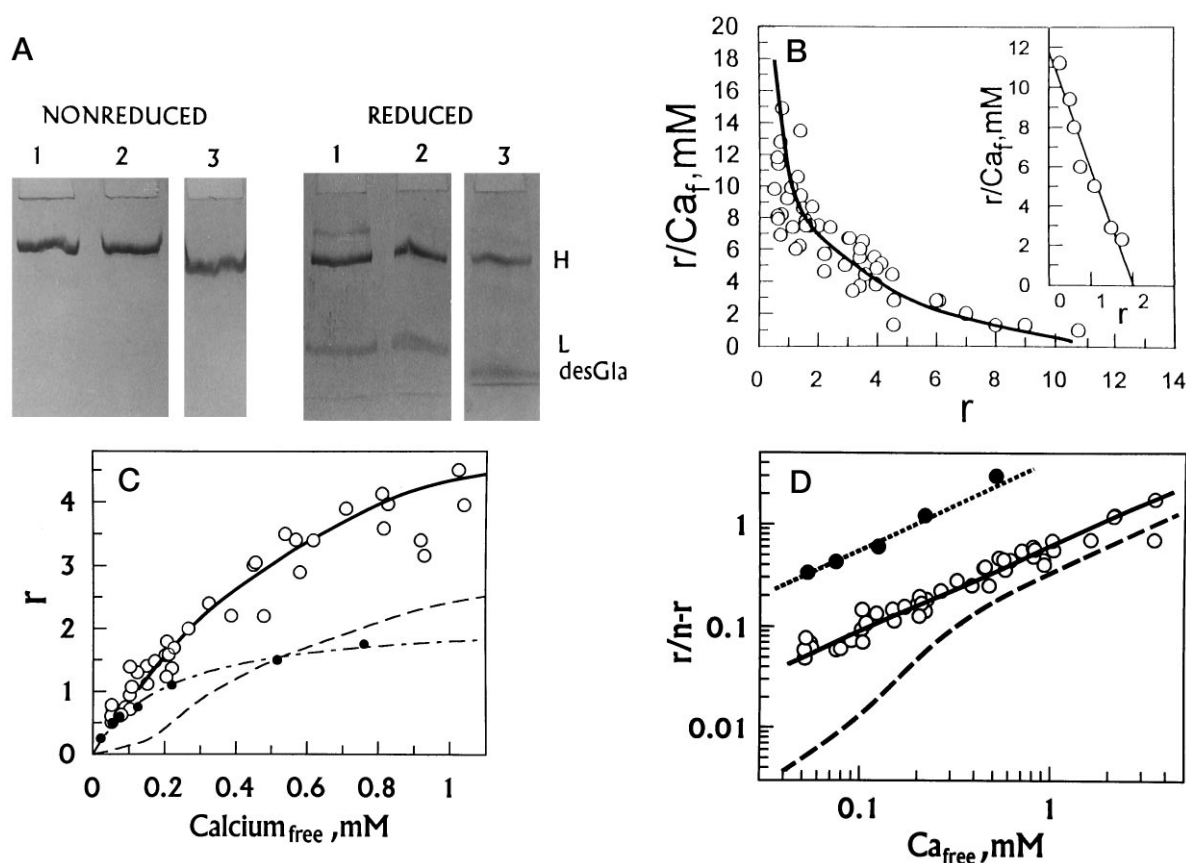


FIG. 1. **Interaction of Ca^{2+} with human factor X.** A, SDS gel electrophoretic analysis of nonreduced (left) and reduced (right) samples of native, decarboxylated, and des-44-X. Gel 1 is of native factor X, gel 2 is of decarboxylated factor X, and gel 3 is of des-44-X. In reduced gels H represents the heavy chain, L represents the light chain, and desGla represents the light chain of factor X from which the Gla domain has been removed. Approximately $4 \mu\text{g}$ of protein was loaded on each lane. B, Scatchard plots of binding of Ca^{2+} to native and decarboxylated human factor X as determined by equilibrium dialysis using ^{45}Ca . The buffer used was TBS, pH 7.4, and the protein concentration in each case was $20 \mu\text{M}$. The range of Ca^{2+} concentrations was $25 \mu\text{M}$ to 5 mM . The data in the main graph are with native factor X, and the data in the inset are with decarboxylated factor X. r , mol of Ca^{2+} bound/mol of protein; Ca_f , free concentration of Ca^{2+} . C, value of r plotted as a function of Ca^{2+} free in solution. To provide an expanded view of the data at lower concentrations of Ca^{2+} , the x axis was cut off at $\sim 1 \text{ mM}$ Ca^{2+} . However, all of the data points were included for obtaining the fitted curves. The open circles represent the data obtained with native factor X, and the closed circles represent the data obtained with decarboxylated factor X. The dashed line without the data legend represents hypothetical binding of Ca^{2+} to the Gla domain and was obtained by subtracting the fitted binding curve of the decarboxylated factor X (●) from that of the native factor X (○). D, Hill plots of Ca^{2+} -binding data. The open circles represent the data for native factor X, and the closed circles represent the data for decarboxylated factor X. The dashed line without the data legend represents hypothetical plot for binding of Ca^{2+} to the Gla domain. n refers to the total number of binding sites, assumed to be 11 for native factor X, 2 for decarboxylated factor X, and 9 for the Gla domain.

TABLE I
Calcium binding characteristics of native, decarboxylated, and des-44-human factor X

The data in Fig. 1 were analyzed using the nonlinear, least squares curve-fitting program LIGAND (20). The number of Ca^{2+} -binding sites represents the average of calcium ions associated with each macromolecule.

Protein	Number of binding sites	K_d
<i>M</i>		
Native factor X		
High affinity sites	2 ± 0.2	$1.0 \pm 0.3 \times 10^{-4}$
Intermediate affinity sites	3.7 ± 0.2	$4.5 \pm 0.7 \times 10^{-4}$
Low affinity sites	5.5 ± 0.3	$2.0 \pm 0.2 \times 10^{-3}$
Decarboxylated factor X	2 ± 0.1	$1.4 \pm 0.2 \times 10^{-4}$
Des-44-X	1 ± 0.2	$1.3 \pm 0.2 \times 10^{-4}$

point of 2 mM Ca^{2+} concentration was 0.89 (Fig. 1D). Since in these calculations we have used nine as the total (intermediate and low affinity) number of binding sites in the Gla domain, the real value of the Hill coefficient at lower concentrations of Ca^{2+} may be higher than 1.57. These data therefore suggest that there are at least two and possibly more cooperative Ca^{2+} -binding sites in the Gla domain of human factor X. This analysis is consistent with the data obtained earlier with bovine factor X and prothrombin (41–44). However, one should note

that our analysis is based upon a hypothetical curve, and additional data will be needed to fully establish the number of sites that exhibit cooperativity in binding of Ca^{2+} to the Gla domain of factor X.

To further investigate the structural requirements for the protease domain binding site, we studied the Ca^{2+} -binding properties of three peptides derived from the catalytic domain of factor X. Peptide 1 containing residues 238–277(58–97) was found to contain a single Ca^{2+} -binding site with a $K_d \sim 150 \mu\text{M}$ (Fig. 2). Ca^{2+} binding to peptide 2 containing residues 238–262(58–82) was weaker ($K_d \sim 600 \mu\text{M}$), and Ca^{2+} binding to peptide 3 containing residues 253–277(73–97) was not observed. In peptide 2, the two residues Asp-250(Glu-70) and Glu-260(80) are included, which participate in the Ca^{2+} binding in the known serine proteases (25, 29, 39). The reason that peptide 2 binds Ca^{2+} with a slightly lower affinity (as compared with peptide 1 and decarboxylated factor X) may be that the β -sheet (β_4 and β_5 strands) that holds these residues together in a loop is not fully formed and is less stable; out of the potential five H-bonds, only three would be possible (Table II). As a result, a significant population of the peptide may not exist in a conformation favorable for Ca^{2+} binding. The binding of Ca^{2+} to the peptide in the favorable conformation will con-

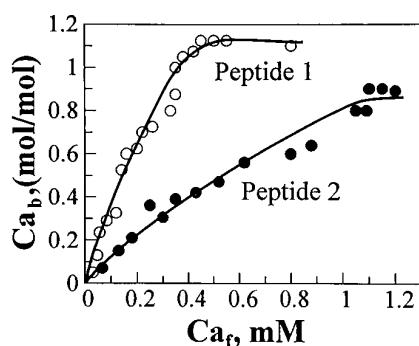


FIG. 2. Binding of Ca^{2+} to peptide 1 corresponding to residues 238–277(58–97) and peptide 2 corresponding to residues 238–262(58–82) of factor X as determined by using a Ca^{2+} -specific electrode. The buffer used was TBS, pH 7.4, and the concentration of each peptide was 400 μM . Similar results were obtained when the concentration of either peptide used was 200 μM . Ca_f and Ca_b refer to free and bound calcium, respectively. $\circ$, peptide 1; $\bullet$, peptide 2.

TABLE II

Hydrogen bonds between the $\beta 4$ and $\beta 5$ strands in the protease domain of human factor Xa

The crystallographic data are from Padmanabhan *et al.* (4). Factor X amino acid numbering system has been used (3). The numbers in parentheses refer to the chymotrypsinogen numbering system. In peptide 2 (residues 238–262(58–82)), the first two H-bonds cannot be formed.

Residue $\beta 4$ strand	Residue $\beta 5$ strand	Distance
		Å
Val-246(66) N	His-263(83) O	2.89
Val-246(66) O	His-263(83) N	2.76
Val-248(68) N	Ala-261(81) O	3.00
Val-248(68) O	Ala-261(81) N	2.88
Gly-249(69) O	Glu-260(80) N	3.05

tinuously shift the equilibrium and generate additional conformers suitable for Ca^{2+} binding. This could explain the reduced affinity of this peptide for Ca^{2+} . Peptide 3, which lacked Asp-250(Glu-70), did not bind Ca^{2+} out to 5 mM. This is consistent with the observation that mutation of Asp-250(70) to lysine abolishes this Ca^{2+} -binding site in factor X (46).

Modeling of the Ca^{2+} -binding Sites— Ca^{2+} -binding sites in the Gla domain of factor X were modeled based upon the structure of the Gla domain of prothrombin in the presence of Ca^{2+} (26) and Sr^{2+} (27). The folding of the Gla domain in this model (Fig. 3) is very similar to the Gla domain of factor VIIa determined by x-ray crystallography (39) and to the modeled structure of protein C (47). Both of these proteins have one residue deletion corresponding to residue 4 in bovine prothrombin. No difficulties were encountered in incorporating four intermediate affinity and four low affinity Ca^{2+} -binding sites in the modeled Gla domain of factor X (Fig. 3). Compared with bovine prothrombin, two additional low affinity sites could be formed involving Gla-32 and Gla-39 in factor X. Ca^{2+} or Sr^{2+} was not found coordinated to Gla-32 (Gla-33 in prothrombin), which is flexibly disordered in the crystal structure of prothrombin fragment 1 (26, 27), and Gla-39 is Ala-40 in bovine prothrombin. In our modeled structure of the Gla domain of human factor X, three hydrophobic residues (Phe-4, Leu-5, and Met-8) point out into the solvent. Based upon the work on other vitamin K-dependent proteins (26, 39, 48–51), these residues could insert into the membrane during prothrombinase assembly. Calcium ions bound to the low affinity sites may also participate in binding to the membrane via phosphate head groups. Quenching of the intrinsic fluorescence observed on Ca^{2+} binding to the Gla domain (52) would appear to be due to the perturbation of the environment of Trp-41 (Fig. 3). Thus, all

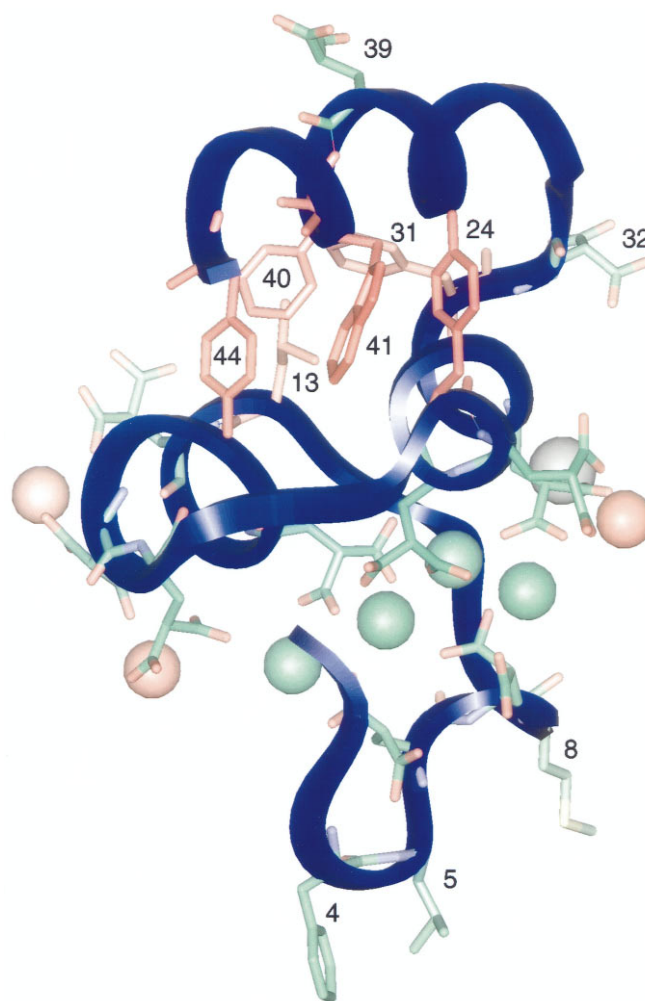


FIG. 3. A modeled structure of the Gla domain of human factor X based upon the structure of the Gla domain of prothrombin in the presence of Ca^{2+} (26) and Sr^{2+} (27). The four green spheres in the center are most likely the intermediate affinity Ca^{2+} -binding sites, and the three red spheres and one black sphere (modeled based upon Sr^{2+} location) represent the low affinity sites. An additional one or two Ca^{2+} -binding sites in factor X may be formed by Gla-32 and/or Gla-39. The side chains of each Gla residue and a few selected other residues are also shown. According to this model, residues 4 (Phe), 5 (Leu), and 8 (Met) as well as Ca^{2+} coordinated to the low affinity sites may directly interact with the PL membrane. Thus, the region up to the array of Ca^{2+} sites may be embedded in the PL membrane. Note the hydrophobic environment of Trp-41 that may be responsible for fluorescence quenching upon Ca^{2+} binding to the Gla domain.

of the known properties of Gla domain of factor X, including the intermediate and low affinity Ca^{2+} -binding sites determined experimentally (Fig. 1) could be accounted for in the modeled structure (Fig. 3).

High affinity Ca^{2+} -binding site in the protease domain of factor X is analogous to the site first observed in trypsin (29). In trypsin, Ca^{2+} -binding site is contained in a surface loop that is formed by residues 249(69)–260(80). In trypsin as well as in factor VIIa and elastase, all of the calcium ligands are provided by the residues in this loop (29, 39, 53). These are the side chain carboxyl oxygens of Asp-250(Glu-70) and Glu-260(80) and the main chain carbonyl oxygens of residues 252(72) and 255(75). Additional ligands may be water molecules or other side chains contained in this so-called calcium binding loop (29, 39, 53). Additionally, this surface loop is stabilized by hydrogen bonds between the carboxyl group of residue 260(80) and the main chain amide groups of residues 258(78) and 259(79). The crystal structure of factor Xa is in the absence of Ca^{2+} , and Glu-

260(80) residue is disordered in this structure (4). However, Glu-260(80) could be easily modeled, and all of the above features observed in the calcium binding loop of trypsin (29) could be readily incorporated into the factor Xa structure. This Ca^{2+} -binding site in the protease domain of factor Xa is shown in Fig. 4.

Functional Significance of the Protease Domain Ca^{2+} -binding Site in Factor Xa— Ca^{2+} potentiated the S-2222 hydrolytic activity of factor Xa β and des-44-Xa β by approximately ~1.6-fold, and the effect was primarily on the k_{cat} (Table III). In three separate experiments, the half-maximal increase was observed at $150 \pm 20 \mu\text{M}$ Ca^{2+} . Since des-44-Xa β does not contain the high affinity site in the EGF1 domain, the Ca^{2+} effects are due to the occupancy of the protease domain Ca^{2+} -binding site. Rezaie and Esmon (46) reported that Ca^{2+} increases ($K_d \sim 250 \mu\text{M}$) the amidolytic activity of a recombinant factor Xa mutant (lacking the Gla and EGF1 domains) toward a synthetic substrate (methoxycarbonyl-D-cyclohexylglycyl-Gly-Arg-*p*-nitroanilide) by 35%. Similarly, Sherrill *et al.* (54) reported that Ca^{2+} ($K_d \sim 200 \mu\text{M}$) enhances the amidolytic activity of native and des-44-Xa by 25–35%. Our data on S-2222 hydrolysis are qualitatively similar to the observations of these authors (46, 54). Importantly, we show that the effect of Ca^{2+} is primarily on the k_{cat} and that the K_m is essentially unaffected (Table III).

In further experiments, we tested whether or not occupancy of Ca^{2+} -binding site in the protease domain prevents proteolytic cleavages in the autolysis loop. Des-44-Xa was incubated at 37 °C in the presence of 1 mM Ca^{2+} or 1 mM EDTA. Samples were removed at different times for SDS gel analysis and for S-2222 activity in the presence or absence of 1 mM Ca^{2+} . These data are presented in Fig. 5. In the presence of Ca^{2+} , the rate of proteolysis in the autolysis loop is ~50% (apparent first order rate constant 0.04/h) of that observed in the presence of EDTA (apparent first order rate constant 0.08/h) as analyzed by the disappearance of H β and appearance of H γ N and H γ C. Additionally, the cleavage(s) in the autolysis loop results in a virtual loss of amidolytic activity. The loss of activity was exponential (Fig. 5), probably as a result of depletion of the substrate and the activator simultaneously. A 30-h preparation of des-44-Xa γ that appeared to be free of des-44-Xa β (as analyzed by SDS gels) retained ~5% of the amidolytic activity of des-44-Xa β . However, our des-44-Xa γ preparation may contain small amounts of des-44-Xa β , which could partly contribute to the observed amidolytic activity. Proteolytic cleavage in our des-44-Xa preparation had occurred at Arg-332(150)–Gln-333(151) (see legend to Fig. 5). Thus proteolysis at the Arg-332(150)–Gln-333(151) peptide bond leads to a loss of catalytic efficiency in factor Xa. Furthermore, Ca^{2+} significantly protects the protease domain from proteolysis in this loop. It should be noted that while our studies were in progress, Prydzial and Kessler (16) reported that factor Xa cleaved at Lys-330(147)–Gly-331(148) by plasmin is devoid of amidolytic and clotting activity. Moreover, thrombin cleaved in the autolysis loop is less stable, and within hours at 37 °C loses its catalytic efficiency (55). Thus, an overall conclusion to be reached from these observations is that proteolysis in the autolysis loop in factor Xa and thrombin leads to an unstable enzyme with concomitant loss of catalytic efficiency.

We next examined whether or not des-44-Xa γ has the ability to bind to *p*-AB. These experiments were designed to test the availability of S1 site in the enzyme. Consistent with earlier observations (54), factor Xa β , decarboxylated Xa β , and des-44-Xa β each were found to interact with *p*-AB with K_d $30 \pm 3 \mu\text{M}$ in both the presence and absence of Ca^{2+} . These observations are also consistent with our earlier findings that the K_m of

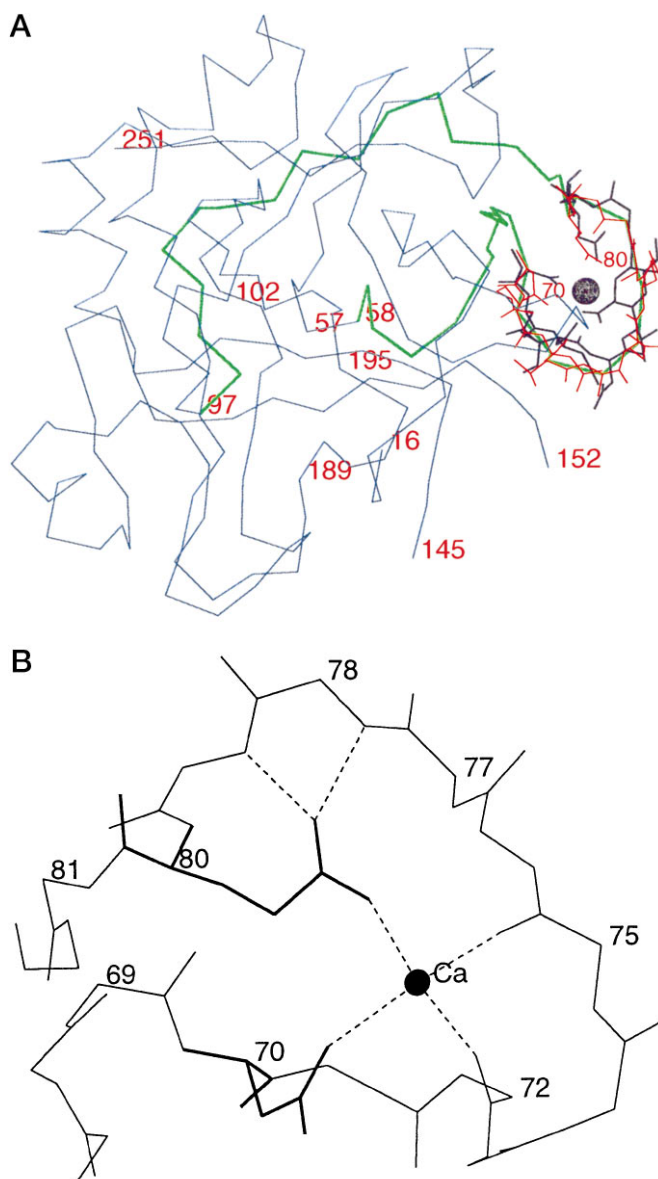


FIG. 4. Location of the Ca^{2+} -binding site in the protease domain of human factor Xa. A, Ca^{2+} tracing of factor Xa protease domain with 250(70) to 260(80) side chains. The structure of factor Xa protease domain is from Padmanabhan *et al.* (4). Blue is the polypeptide Ca^{2+} tracing for factor Xa protease domain, purple is 70–80 residues of trypsin superimposed on the factor Xa structure, red is the 250(70) to 260(80) residues of factor Xa, and green is the peptide of residues 238(58) to 277(97) used for the Ca^{2+} -binding experiment (Fig. 2). The Ca^{2+} is shown as a purple sphere on the top right involving Asp-250(Glu-70) and Glu-260(80) residues. The sequence 328(145)–334(152) represents the autolysis loop. Note that there is a one-residue deletion in factor X compared with chymotrypsinogen in this loop (4, 25). Location of the catalytic triad residues His-236(57), Asp-282(102), and Ser-379(195) as well as S1 residue Asp-373(189) is also shown. B, enlarged view of the Ca^{2+} -binding loop. Backbone atoms along with the side chains of Asp-250(70) and Glu-260(80) are shown. Ca^{2+} is shown as a solid circle, and its coordination with the carbonyl oxygens of residues 252(72) and 255(75) and carboxyl oxygens of Asp-250(70) and Glu-260(80) are indicated by dashed lines. Hydrogen bonds formed between the carboxyl group of Glu-80 and the main chain amide groups of residues 78 and 79 are also indicated by dashed lines. The average root mean square deviation of Ca^{2+} atoms of residues 70–80 between the modeled loop (plus Ca^{2+}) and the observed structure (in the absence of Ca^{2+}) of the loop is 0.53 Å. The chymotrypsinogen numbering is used in the figure.

S-2222 hydrolysis by factor Xa is not affected by Ca^{2+} (Table III). Des-44-Xa γ also interacted with *p*-AB, although with slightly reduced affinity (K_d $40 \pm 5 \mu\text{M}$); however, the enhance-

TABLE III
Effect of calcium on the hydrolysis of S-2222 by native and des-44-Xa β in TBS, pH 7.4 buffer

The results presented are average of three experiments.

	Native factor Xa β		Des-44-factor Xa β	
	K_m	k_{cat}	K_m	k_{cat}
	μM	s^{-1}	μM	s^{-1}
Ca $^{2+}$ (1 mM)	170 $\pm$ 18	168 $\pm$ 12	181 $\pm$ 21	154 $\pm$ 19
EDTA (1 mM)	159 $\pm$ 22	103 $\pm$ 15	165 $\pm$ 23	95 $\pm$ 18

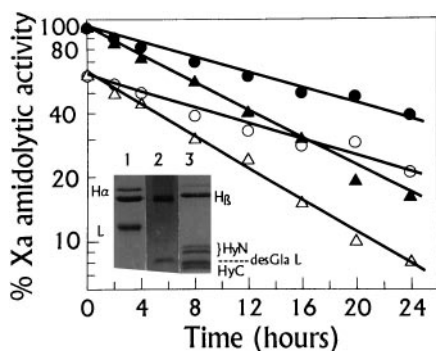


FIG. 5. Effect of Ca $^{2+}$ on the proteolytic cleavage in the autolysis loop of des-44-Xa. In one experiment, the protein in TBS, pH 7.4, was incubated in the presence of 1 mM Ca $^{2+}$ and assayed for S-2222 activity (170 μM) in the presence of 1 mM Ca $^{2+}$ (●) or 1 mM EDTA (○). In the other experiment, the protein was incubated in the presence of 1 mM EDTA and assayed for S-2222 activity in the presence of 1 mM Ca $^{2+}$ (▲) or 1 mM EDTA (△). The samples were also removed and analyzed by SDS gel electrophoresis. The gel 2 in the inset is des-44-Xa, and gel 3 is des-44-Xa incubated for 8 h in the presence of EDTA. Gel 1 is of native factor Xa and is shown here for comparison. The proteins in gel 3 (a separate gel was run with five lanes) were transferred to polyvinylidene difluoride membrane and sequenced. H γ N corresponded to sequence of the NH $_2$ -terminal of heavy chain (IVGGQ), whereas the desGla L corresponded to the light chain (KDGDQ) and the H γ C corresponded to the autolyzed newly formed COOH fragment of heavy chain (QSTRL). Single-letter codes for the amino acids have been used.

ment in the intrinsic fluorescence of *p*-AB upon binding at the active site was only ~25% of that observed with des-44-Xa β . It should be noted that an enhancement of the intrinsic fluorescence of *p*-AB was not observed when DEGR-des-44-Xa γ (or DEGR-des-44-Xa β) was used in the *p*-AB titration experiments; this strongly indicates that the increase in intrinsic fluorescence observed with des-44-Xa γ is due to the binding of *p*-AB at the active site. Since a reduced fluorescence increase was observed with des-44-Xa γ , it would indicate that the environment of the *p*-AB bound to the S1 site in des-44-Xa γ is less nonpolar as compared with that in the des-44-Xa β molecule. Based upon the work with other serine proteases (4, 56), it would appear that Trp-399(215) in factor Xa contributes to the nonpolar environment and therefore enhancement of *p*-AB intrinsic fluorescence upon binding at the active site; if so, then this region in factor Xa γ is slightly perturbed without the loss of S1 binding site.

In the following experiments, we wished to investigate the effect of proteolysis in the autolysis loop of factor Xa β in its binding to factor Va. The rationale for these experiments is based upon the knowledge that the protease domain of factor Xa may be involved in binding to factor Va (38, 57). For these studies, we prepared DEGR-Xa β , decarboxylated DEGR-Xa β , and DEGR-Xa $\beta\gamma$ (a mixture containing 50% of DEGR-Xa β and 50% of DEGR-Xa γ , see Experimental Procedures and legend to Fig. 6), and evaluated their abilities to compete with active factor Xa β in binding to factor Va in the prothrombinase complex. We also evaluated the abilities of the three peptides to compete for factor Va in the prothrombinase complex. First, we determined the functional K_d (EC $_{50}$) of interaction of factor Xa β

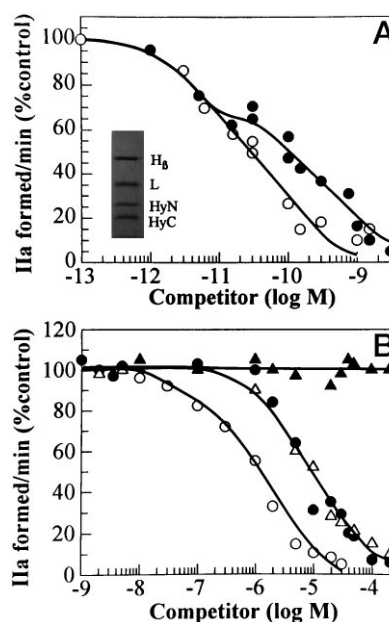


FIG. 6. Abilities of the active-site blocked factor Xa molecules and three peptides containing the factor X sequence to inhibit thrombin generation in a Xa/Va/PL system. A system consisting of factor Va (5 μM), factor Xa β (15 μM), PCPS vesicles (10 μM), and prothrombin (750 nM) in TBS/BSA, pH 7.4, containing 5 mM Ca $^{2+}$ was used to study the abilities of the competitors to inhibit prothrombin activation. The results of thrombin (IIa) generation are expressed as the percentage of the value obtained with factor Xa β alone (i.e. percent control); the rate of IIa generation under these conditions was 0.6 nM/min, which is comparable to an expected rate (58). Increasing concentrations of the competitor were mixed with a fixed (15 μM) concentration of factor Xa β prior to incubation with factor Va and PCPS vesicles. Factor Va subunits were associated as outlined (59). Factor Va (6.3 mg/ml, 37.5 μM) obtained in 50% glycerol, 2 mM CaCl $_2$ was diluted 100-fold in TBS/BSA/2 mM CaCl $_2$, pH 7.4, and incubated for 2 h at 37 °C. The diluted sample was kept at 4 °C and used within 24 h. The data presented are the average of two experiments. A, abilities of DEGR-Xa β (○) and DEGR-Xa $\beta\gamma$ (●) to inhibit binding of factor Xa β to factor Va in the prothrombinase complex. Factor Xa $\beta\gamma$ preparation prior to DEGR-CK incorporation had 50 $\pm$ 5% of the S-2222 hydrolytic activity of factor Xa β at similar concentrations. The inset shows the reduced SDS gel of DEGR-Xa $\beta\gamma$ sample used in this experiment. B, abilities of decarboxylated DEGR-Xa β , peptide 1 (residues 238–277(58–97)), peptide 2 (residues 238–262(58–82)), and peptide 3 (residues 253–277(83–97)) to inhibit the binding of factor Xa β to factor Va in the prothrombinase complex. ○, decarboxylated DEGR-Xa β ; ●, peptide 1; ▲, peptide 2; and △, peptide 3. Finally, it should be noted that none of the competitors inhibit prothrombin activation in a Ca $^{2+}$ /PL system (i.e. without factor Va).

with factor Va in our system. A K_d value of 30 $\pm$ 5 μM was obtained for this interaction and its value was not dependent upon the concentration of prothrombin utilized in the assay (see "Experimental Procedures"). Next, using this value of functional K_d and steady state inhibition curves, the K_d values for interaction of factor Va with various competitors were calculated. These data are presented in Fig. 6. Using the data in this figure, a K_d value of 26 $\pm$ 4 μM for the interaction of DEGR-Xa β and factor Va was calculated. This agrees well with the functional K_d value of 30 $\pm$ 5 μM calculated from the initial rates of prothrombin activation assays using different concentrations of factor Xa β and a fixed (5 μM) concentration of factor Va (see above). When DEGR-Xa $\beta\gamma$ (50% each of β and γ species; see legend to Fig. 6) was used as a competitor, the inhibition curve best fitted ($p < 0.01$) to a two-site model, with a K_d value of 19 $\pm$ 3 μM and 231 $\pm$ 15 μM . The value of 19 μM was taken as the K_d for the interaction of factor Va and DEGR-Xa β , and the value of 231 μM was taken as the K_d for the interaction of factor Va and DEGR-Xa γ . Decarboxylated DEGR-Xa β interacted with factor Va with $K_d \sim 1.5 \pm 0.5 \mu M$, and peptides 1 and 3

interacted with factor Va with $K_d \sim 10 \pm 2 \mu\text{M}$, whereas peptide 2 did not bind to factor Va.⁵

Our observation that the interaction of factor Va and factor Xa on the PCPS vesicles is not influenced by the substrate concentration in the prothrombinase assay system is in agreement with the earlier data of Nesheim *et al.* (60). Our K_d value of 30 pM for the association of factor Xa β and factor Va on the phospholipid vesicles is also in agreement with the value of 25 pM recently obtained by Ye and Esmon (61). Furthermore, our K_d value of 1.5 μM for the interaction of decarboxylated DEGR-Xa β (which cannot bind to PL vesicles) and factor Va also agrees well with previous determinations on the K_d of interaction of factors Va and Xa in the absence of phospholipid using physical methods (62). Finally, our peptide inhibition data are also in general agreement with the data of Chattopadhyay *et al.* (63), who observed that a peptide containing the sequence 263(81)-274(94) of factor Xa inhibited prothrombin activation with an IC_{50} (50% inhibition) value of 20 μM . The above comparisons attest to the validity of our system in studying Xa:Va interactions. Using this system, we show that as compared with factor Xa β , factor Xa γ binds to factor Va with ~ 10 -fold reduced affinity. Overall, our data reveal that factor Xa γ is essentially devoid of catalytic activity and has a reduced affinity for factor Va; however, the S1 binding site is retained in this molecule. Similarly, in another study (65), we show that factor IXa cleaved in the autolysis loop retains the S1 site, has reduced affinity for factor VIIIa and is devoid of enzymatic activity.

Concluding Remarks—Collectively, our data combined with previous observations support the conclusion that factor X contains one high affinity Ca^{2+} -binding site in the protease domain and one in the EGF1 domain. The formation of the high affinity site in the EGF1 domain requires the presence of the Gla domain, which could be in the decarboxylated form. The Gla domain contains four intermediate and five to six low affinity Ca^{2+} -binding sites, which could be successfully modeled based upon the crystallographically determined structure of the Gla domain of prothrombin (26, 27). The protease domain Ca^{2+} -binding site could also be easily modeled into the x-ray structure of factor Xa (4). Using the x-ray structure of EGF2 and protease domains of factor Xa (4), the NMR coordinates of bovine factor X EGF1 domain (28), and x-ray coordinates of the Gla domain of bovine prothrombin (26, 27), we were able to model the entire factor Xa molecule. This is shown in Fig. 7. The model of the whole factor Xa molecule is shown here for clarity in pointing out the location of the Ca^{2+} -binding and membrane insertion sites in the protein. However, it should be borne in mind that it is a modeled view of the protein, and it is depicted here simply to illustrate the biologic features. Biochemical data support the conclusion that the protease domain of factor Xa is important in its interaction with factor Va and that the Gla domain is needed for insertion into the PL membrane. Based upon peptide inhibition data, the region of the protease domain containing the peptide sequence 263(81)-274(94)(63) may be important for Xa:Va interaction. However,

⁵ In the absence of factor Va, none of the three peptides inhibited prothrombin activation. The protocol for these experiments was the same as described under "Experimental Procedures" in the presence of factor Va, and the data presented are the average of two experiments. The rate of thrombin generation was $5 \pm 0.3 \text{ nm/min}$ in a system containing 10 nM factor Xa, 5 mM Ca^{2+} , 10 μM PL, and 700 nM prothrombin. In separate experiments, inclusion of 40 μM of each peptide in the reaction mixture gave the following initial rates of thrombin generation. Peptide 1, $4.7 \pm 0.2 \text{ nm/min}$; peptide 2, $4.9 \pm 0.3 \text{ nm/min}$; peptide 3, $4.6 \pm 0.2 \text{ nm/min}$. Thus, it would appear that the inhibition observed with peptides 1 and 3 in the presence of factor Va is not a consequence of their utilization as substrates in the chromogenic assay used to measure thrombin activity.



FIG. 7. A ribbon model structure of human factor Xa. Purple represents Gla domain, green represents EGF1 domain, orange represents EGF2 domain, and cyan represents protease domain. The active site serine and the protease domain Ca^{2+} -binding loop are shown in red. The four green spheres represent intermediate affinity and one black and three red spheres represent low affinity Ca^{2+} -binding sites in the Gla domain. The low affinity Ca^{2+} -binding sites and the three hydrophobic residues (green) protruding downward are thought to participate in binding to the PL membrane (see Fig. 3). The Ca^{2+} -binding sites in the EGF1 and the protease domains are shown as purple spheres. In this model, relative orientations of the domains with respect to each other were positioned based upon the crystal structure of porcine factor IXa (30). Another constraint in building this model was to maintain a distance of $\sim 70 \text{ \AA}$ between the array of seven calcium ions of the Gla domain and the active site serine in the protease domain; this distance is approximated from the fluorescence energy transfer experiments (65).

one should note that a peptide may adopt a different conformation than that occurring in the protein. Thus, additional data will be required to confirm whether or not this segment of the protease domain interacts with factor Va. The precise functions of the EGF1 and EGF2 domains are not as yet clear, but may be involved in protein-protein or protein-cofactor interactions. Finally, Ca^{2+} binding to the protease domain protects proteolysis in the autolysis loop, and stabilizes this domain for optimal catalytic activity and factor Va binding.

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